



GBC1, WS 2025

Chlorophyll Determination

Determination of Chlorophyll Concentration in Biomatter using Spectrophotometry

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1 Background

1.1 Chlorophyll

Chlorophyll is a green pigment found in plants and some bacteria. It is responsible for the green color of plants and is essential for photosynthesis. It consists of two main types: Chlorophyll a (Chl_a) and Chlorophyll b (Chl_b).

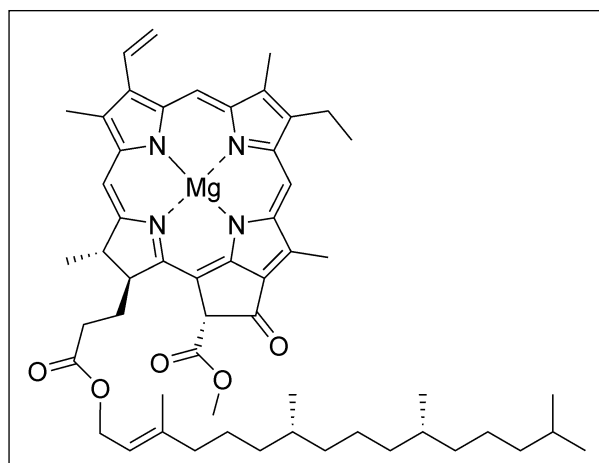


Figure 1: Chlorophyll A structure ^[1]

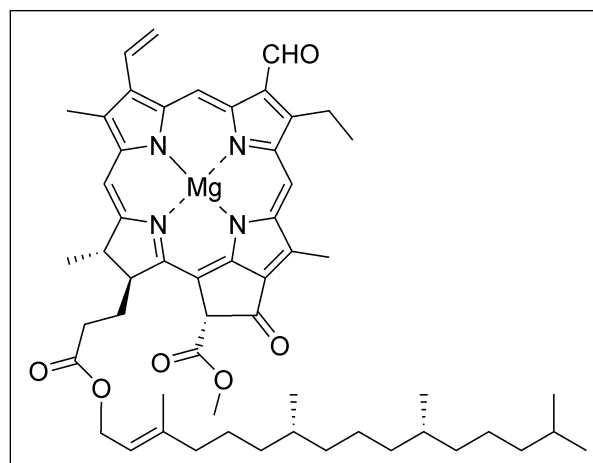


Figure 2: Chlorophyll B structure ^[2]

1.1.1 Solubility

Chlorophyll has a long fatty chain called phytol ($\text{C}_{20}\text{H}_{39}$) attached to it. This chain makes chlorophyll dissolve well in oils and organic solvents, but not in water. The phytol chain acts like an anchor that holds the chlorophyll in place within the plant's membrane structures.

Because of this fatty chain, organic solvents like acetone, methanol, or ethanol are needed to extract chlorophyll from plant tissues. 80% acetone (mixed with 20% water) is commonly used because the small amount of water helps break apart the protein structures that hold the chlorophyll. ^{[3], [4]} 100% acetone is very suitable for leave extracts. ^[3]

1.1.2 Solvatochromism

The solvent effect, known as "Solvatochromism", is a phenomenon that occurs when the absorption spectrum of a compound changes between different solvents used. This solvent-dependent effect applies to chlorophyll and arises from the interaction between the solvent molecules and the chlorophyll's π -electron cloud ^{[5], [6]}.

Consequently, absorption coefficients derived for methanol cannot be used for acetone extracts, and coefficients for 100% acetone cannot be used for 80% acetone. The specific equations discussed in Section 5 (using 11.78 and 2.29)¹ are strictly valid only for 80% Acetone extracts. ^{[3], [5]}

1.2 Spectrophotometry

1.2.1 Beer-Lambert law

The Beer-Lambert law, shown in Equation 1, describes the correlation between the dimensionless total absorbance E_λ , the molar extinction coefficient ϵ_λ ($\frac{\text{l}}{\text{mol} \cdot \text{cm}}$ ^[7]) and concentration c ($\frac{\text{mol}}{\text{l}}$) of

¹replace with actual section used in this document

the substance of interest in the sample, and the light path length d (cm) through the sample, where λ is an arbitrary wavelength of light.

$$(1) \quad E_{\lambda} = \varepsilon_{\lambda} \cdot c \cdot d$$

For example Chl_a has $\varepsilon_{420} = \frac{85\,000}{\text{mol} \cdot \text{l}}$. This however depends on a multitude of factors, such as pH, temperature or solvent, and is therefore not a constant [7].

1.2.2 Calculation of Chlorophyll Concentration

By rearranging Equation 1 we get:

$$(2) \quad c \left[\frac{\text{mol}}{\text{l}} \right] = \frac{E_{\lambda}}{\varepsilon_{\lambda}} \cdot \frac{1}{d}$$

d is the path length through the sample or in other words the thickness of the cuvette. The global standard for this is $d = 1\text{ cm}$ [8]. Therefore we can effectively cancel out d in the equation.

While Equation 2 is a general equation yielding $\frac{\text{mol}}{\text{l}}$, we can convert to mass concentration $\frac{\text{g}}{\text{l}}$ by multiplying with the molar mass MG_S of our substance S and factoring in the dilution factor $\frac{V_E}{V_P}$, where V_E is the volume of the extraction solvent and V_P is the volume of the biological source material.

$$(3) \quad c \left[\frac{\text{g}}{\text{l}_E} \right] = \frac{E_{\lambda}}{\varepsilon_{\lambda}} \cdot \text{MG}_S \cdot \frac{1}{d} \Rightarrow$$

$$c \left[\frac{\text{g}}{\text{l}_P} \right] = c \left[\frac{\text{g}}{\text{l}_E} \right] \cdot \frac{V_E}{V_P}$$

Here $c \left[\frac{\text{g}}{\text{l}_E} \right]$ is the mass concentration of the substance S in the extraction solvent and $c \left[\frac{\text{g}}{\text{l}_P} \right]$ is the mass concentration of the substance S in the biological source material.

2 Execution

3 Results

We performed various analysis tasks in order to gain a better understanding of the data and the results. These were done in isolation to avoid any confounding factors.

3.1 Spectral Analysis

We decided to take a detailed full spectrum sample using ...². The full spectrum data was derived by sampling absorption rates at 10nm intervals from the spectrophotometer's automatic full spectrum curve. Results are shown in Figure 4.

3.1.1 Reference spectrum

We compare our full spectrum data with the reference spectra of chlorophyll a and chlorophyll b in diethyl ether. This is used for reference even though we used acetone in our experiment, because we failed to find other publically available full spectrum reference data.

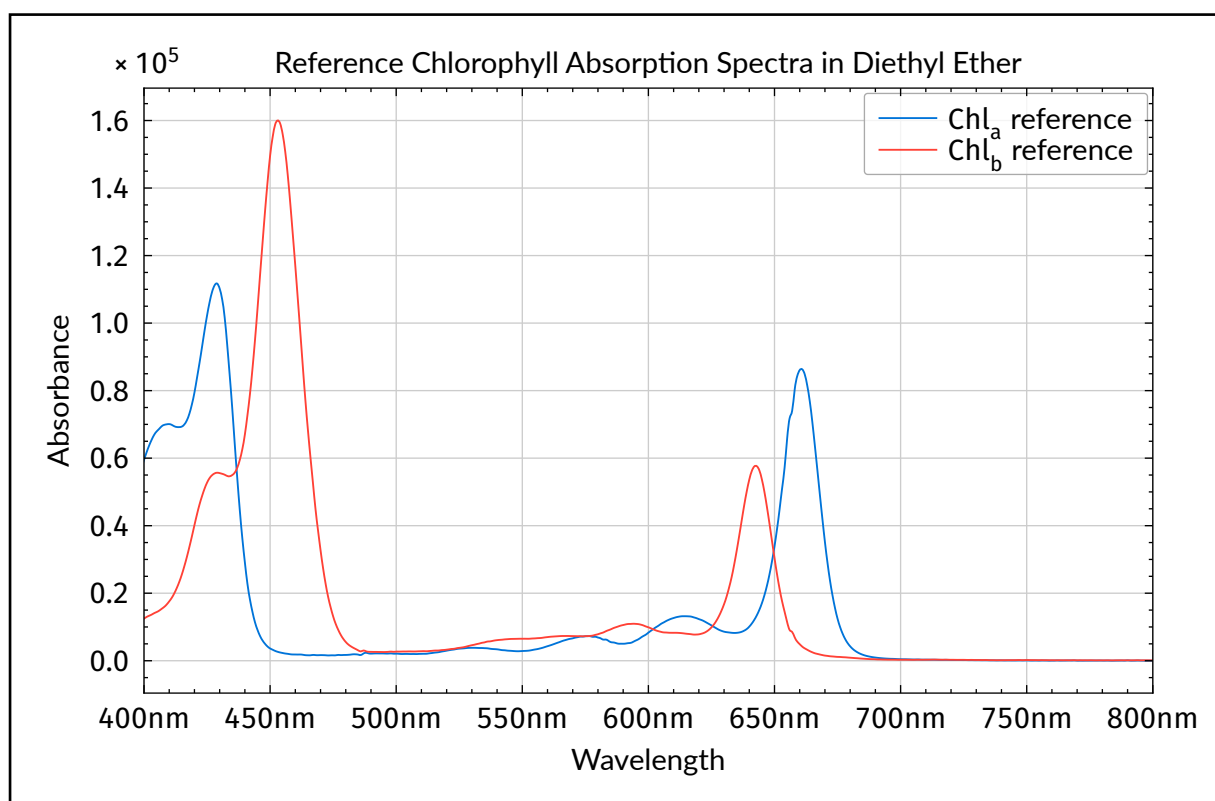


Figure 3: Reference Chlorophyll Absorption Spectra in Diethyl Ether^{[1], [2]}

Blue shows Chl_a and red shows Chl_b

In Figure 3 we can observe the blue spectrum peaks at ~ 430nm for Chl_a and ~ 452nm for Chl_b. The red spectrum peaks at ~ 660nm for Chl_a and ~ 642nm for Chl_b.

These match the characteristics of the chlorophylls absorption spectrum in diethyl ether. We can observe the spectrum line of Chl_a buckles when crossing the line of Chl_b in the blue region, which is also characteristic for both chlorophyll variants.^[5]

Therefore, we conclude the found dataset to be a viable and accurate source of reference.

²hier einfügen von welcher Gruppe und Sample aus Excel Datei online

3.1.2 Sample spectrum

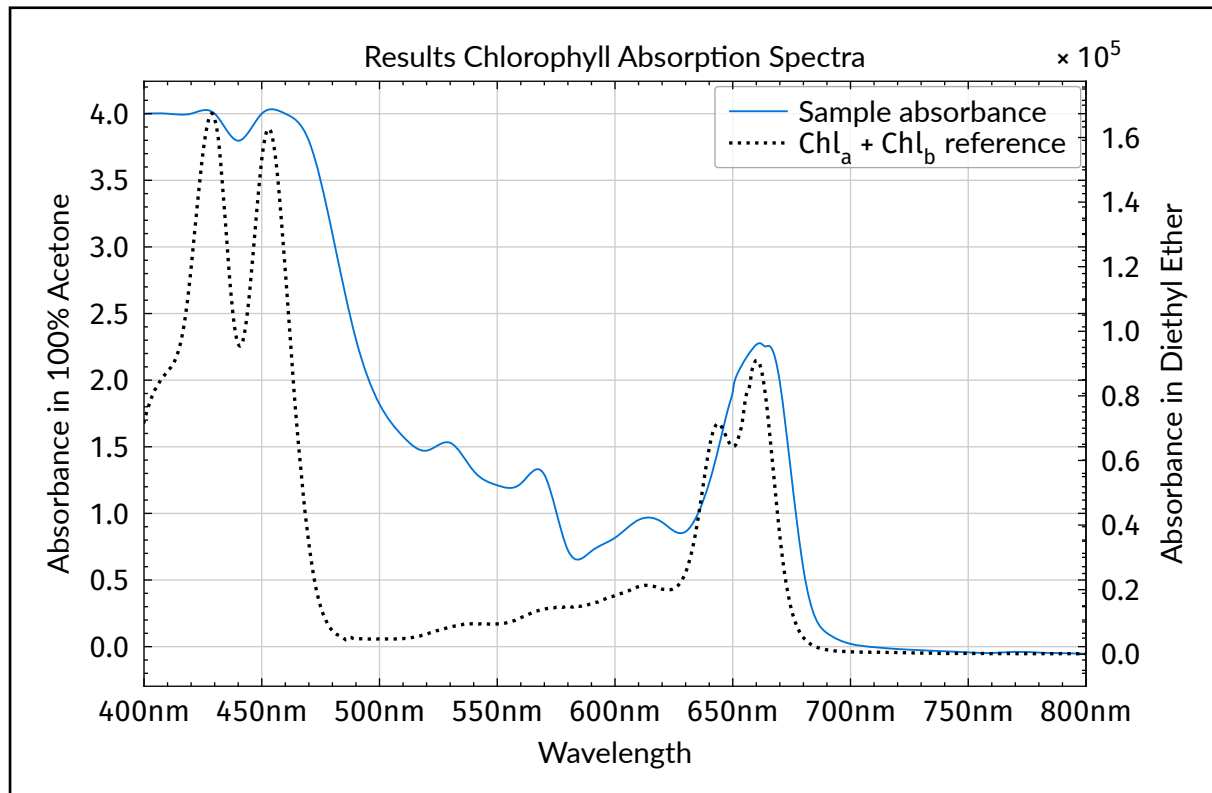


Figure 4: Absorption Spectrum of the Sample (shown in blue) and combined $\text{Chl}_a + \text{Chl}_b$ Reference from Figure 3 (shown in dotted black)

The sample spectrum shows the absorption of the total, non-isolated chlorophylls ($\text{Chl}_a + \text{Chl}_b$) in the sample. We can observe three peaks: two in the blue region at $\sim 430\text{nm}$ and $\sim 445\text{nm}$ with one in the red region at $\sim 660\text{nm}$.

We can observe similar peaks as in the reference spectrum, but with slight distortions and more or less pronounced features. This is because the sample consists of total, non-isolated chlorophylls ($\text{Chl}_a + \text{Chl}_b$) and possibly other pigments dissolved in acetone. Furthermore, as mentioned in Section 1.1.2, shifts are to be expected when comparing our sample with diethyl ether dissolved reference^[5]. Despite this, one can even spot fine features in our spectrum. For example the valley at the $\sim 650\text{nm}$ wavelength that is clearly visible in the reference spectrum is present in our data in form of a small bump.

The most claring difference compared to reference is the samples gradually increasing absorption in the green spectrum (read from right/red to left/blue), whereas according to the reference the absorption should slightly decrease. One possible explanation for this is the presence of other pigments in the sample that absorb in the green spectrum, such as carotenoids^{[4], [5]}. Another plausible explanation is incomplete calibration: Calibration did not happen across the entire spectrum but only once on the higher end of the wavelength. The level of unfamiliarity with the spectrophotometer and the spectrometer itself might have contributed to this.

3.2 Chlorophyll Concentration with Given Formula

The instructor provided us with a formula to calculate the chlorophyll concentration:

$$\begin{aligned}
 (4) \quad c_a \left[\frac{\text{mg}}{\text{l}} \right] &= 11.78 E_{664} - 2.29 E_{647} \\
 c_b \left[\frac{\text{mg}}{\text{l}} \right] &= 20.05 E_{647} - 4.77 E_{664} \\
 c_{\text{total}} &= c_a + c_b = 27.8 E_{652}
 \end{aligned}$$

Plugging in the concrete values, shown in Table 1, for our sample we get:

$$\begin{aligned}
 c_a \left[\frac{\text{mg}}{\text{l}} \right] &= 11.78 \cdot 2.253 - 2.29 \cdot 1.715 = \mathbf{22.61299} \\
 c_b \left[\frac{\text{mg}}{\text{l}} \right] &= 20.05 \cdot 1.715 - 4.77 \cdot 2.253 = \mathbf{23.63894} \\
 c_{\text{total}} \left[\frac{\text{mg}}{\text{l}} \right] &= c_a + c_b = 22.61299 + 23.63894 = \mathbf{46.25193} \\
 &= 27.8 E_{652} = 27.8 \cdot 2.049 = \mathbf{56.9622}
 \end{aligned}$$

Wavelength	Absorbance
664	2.253
647	1.715
652	2.049

Table 1: Wavelengths and absorbance of the results from our sample needed for calculating the chlorophyll concentration using the given formulas

A quick sanity check with the Gemini LLM for the plausability of these values was positive. Lab extracts have a common range of $10 - 30 \frac{\text{mg}}{\text{l}}$. This aligns with our values.

We can see a significant difference between the total result using the standalone absorbance factor and the addition of Chl_a and Chl_b , with an absolute difference of $10.7103 \frac{\text{mg}}{\text{l}}$ and a relative difference of 18.8 %.

3.2.1 Interpretation of the difference

It proves difficult to reason for a possible explanation for this discrepancy without knowing where the given formulas originate from and how they were derived. Measurement or execution errors are a possibility, even though seeming unlikely due to the high difference.

3.3 Chlorophyll Concentration using Beer-Lambert law

In order to find both the Chl_a and Chl_b concentration, we need to use the Beer-Lambert law. We arrange the Equation 1 into a linear equation system to solve for both concentrations:

$$\begin{aligned}
 (5) \quad E_{664} &= \varepsilon_{a,664} \cdot c_a + \varepsilon_{b,664} \cdot c_b \\
 E_{647} &= \varepsilon_{a,647} \cdot c_a + \varepsilon_{b,647} \cdot c_b
 \end{aligned}$$

Because we use the standard path length of $d = 1\text{cm}$ we can ignore $\frac{1}{d}$ in our calculations. Using substitution we can solve for c_a and c_b :

$$(6) \quad c_a \left[\frac{\text{mol}}{\text{l}} \right] = \frac{E_{664} \varepsilon_{b,647} - E_{647} \varepsilon_{a,664}}{\varepsilon_{a,664} \varepsilon_{b,647} - \varepsilon_{a,647} \varepsilon_{b,664}}$$

$$c_b \left[\frac{\text{mol}}{\text{l}} \right] = \frac{E_{647} \varepsilon_{a,664} - E_{664} \varepsilon_{b,647}}{\varepsilon_{a,664} \varepsilon_{b,647} - \varepsilon_{b,664} \varepsilon_{a,647}}$$

We arrive at final formula for calculating the total chlorophyll concentration, accounting for the mass concentration and dilution factor:

$$(7) \quad c_a \left[\frac{\text{g}}{\text{l}_E} \right] = c_a \left[\frac{\text{mol}}{\text{l}} \right] \cdot \text{MG}_{\text{Chl}_a}$$

$$c_b \left[\frac{\text{g}}{\text{l}_E} \right] = c_b \left[\frac{\text{mol}}{\text{l}} \right] \cdot \text{MG}_{\text{Chl}_b}$$

$$c_{\text{total}} \left[\frac{\text{g}}{\text{l}_E} \right] = c_a \left[\frac{\text{g}}{\text{l}_E} \right] + c_b \left[\frac{\text{g}}{\text{l}_E} \right]$$

$$c_{\text{total}} \left[\frac{\text{g}}{\text{l}_P} \right] = c_{\text{total}} \left[\frac{\text{g}}{\text{l}_E} \right] \cdot \frac{V_E}{V_P}$$

By plugging in the values from Table 2 for this equation we arrive at the final values for the mass concentration in the extraction solvent:

$$c_a \left[\frac{\text{mg}}{\text{l}_E} \right] = \mathbf{23.38}$$

$$c_b \left[\frac{\text{mg}}{\text{l}_E} \right] = \mathbf{24.27}$$

$$c_{\text{total}} \left[\frac{\text{mg}}{\text{l}_E} \right] = \mathbf{47.65}$$

Accounted for the dilution factor from the source material, the final determined mass concentration in the biological source material is:

$$c_{\text{total}} \left[\frac{\text{mg}}{\text{l}_P} \right] = c_{\text{total}} \left[\frac{\text{mg}}{\text{l}_E} \right] \cdot \frac{V_E}{V_P} = \mathbf{190.59}$$

3.3.1 Comparison to given formula

The values calculated using the Beer-Lambert law are in line with the values calculated using the given formulas in Section 3.2. The difference is small (2.93 %) and indicates that the Beer-Lambert law and our calculations are a valid method for determining the chlorophyll concentration. Small differences can be attributed to different extinction coefficients of the chlorophylls used when precalculating the given formulas' factors.

Section 3.2.1 describes the unknown origin of the instructor-provided formulas. After performing the calculations ourselves, we can observe that once the extinction coefficients ε and

	MG_S	ε_{647}	ε_{664}	E_{647}	E_{664}	$V_E \div V_P$
Chl_a	893.5 g mol ⁻¹	17 464 l mol ⁻¹ cm ⁻¹	76 790 l mol ⁻¹ cm ⁻¹	1.715	2.253	$\frac{20\text{ml}}{5\text{ml}} = 4$
Chl_b	907.5 g mol ⁻¹	47 040 l mol ⁻¹ cm ⁻¹	9 121 l mol ⁻¹ cm ⁻¹			

Table 2: Extinction coefficients ^[9], molecular weights, absorbance values and dilution factor needed for calculating the chlorophyll concentration using the Beer-Lambert law.

molecular weights M_G are known for both chlorophyll types, the concentration equation reduces to simple multilinear functions with constant factors applied to the remaining variables E_λ . This suggests that the given formulas are simplified versions of the complete calculation chain we performed based on the Beer-Lambert law.

3.4 Descriptive Statistics

4 Hemoglobin

Hemoglobin Hb is a vital oxygen-carrying protein found in red blood cells. It consists of four subunits:

- 2x α -chains
- 2x β -chains

Each chain contains a heme group with a central iron ion in the ferrous state Fe^{2+} , which is responsible for binding oxygen.

4.1 Methemoglobin

When the iron in Hb is oxidized to the ferric state Fe^{3+} , it forms methemoglobin MetHb, which cannot bind oxygen and therefore cannot participate in oxygen transport.

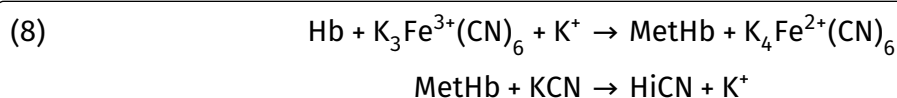
The oxidation of Hb to MetHb is a normal process, but the human body tries to keep the amount of MetHb at < 1% of the total Hb. Enzymes in red blood cells (like NADPH methemoglobin reductase) continuously reduce the iron back to Fe^{2+} to regenerate functional hemoglobin. However, if someone is exposed to certain oxidizing agents (for example, nitrates in well water, benzocaine, or dapsone), the rate of $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ conversion can overwhelm the repair enzymes.

The result is methemoglobinemia, where a significant fraction of hemoglobin is stuck in the Fe^{3+} form. Blood with high methemoglobin turns a chocolate-brown color and can cause symptoms of hypoxia (like cyanosis, fatigue or even neurological symptoms at very high levels)

4.2 Hemiglobincyanide Method

To measure the concentration of hemoglobin in blood samples, one reliable laboratory method is the hemiglobincyanide HiCN method, also known as the cyanmethemoglobin method. This technique converts all forms of hemoglobin (except sulfhemoglobin) into a single, stable colored complex, cyanmethemoglobin = hemiglobincyanide, which can be measured photometrically. This approach is recommended by the World Health Organization (WHO) due to its high accuracy and reproducibility.

The conversion is achieved using Drabkin's reagent, which contains potassium ferricyanide $\text{K}_3\text{Fe}(\text{CN})_6$ and potassium cyanide KCN. Equation 8 describes: The ferricyanide oxidizes hemoglobin to MetHb, and the cyanide then binds to MetHb to form HiCN.



The resulting HiCN complex is stable and has a specific absorbance peak at wavelengths around 540 – 546nm. This property allows for photometric measurement of hemoglobin concentration

using a spectrophotometer. Finally the hemoglobin concentration is determined based on the Beer–Lambert law, shown in Section 1.2.1.

(9)

$$A = \varepsilon * c * l$$

A... total absorbance

ε ... molar absorptivity of the substance

c... concentration of substance in the sample

l... path length through sample

4.3 Reflotron Method

An alternative method for hemoglobin determination is the Reflotron system, a point-of-care device that uses dry reagent strips and reflectance photometry. A drop of whole blood is applied to a test strip, and the device measures the color intensity reflected from the strip to calculate hemoglobin concentration. While the Reflotron provides rapid and user-friendly results, the HiCN method remains the gold standard due to its accuracy and standardization.

5 Procedure

Due to unfortunate circumstances, the Reflotron during our lab session was defective and therefore no test data using that method could be collected. As a result, this chapter only covers the hemiglobincyanide method.

5.1 Materials

- Fresh blood sample (capillary or venous)
- Drabkin's reagent (contains $\text{K}_3\text{Fe}(\text{CN})_6$ and KCN)
- Spectrophotometer with a 546 nm filter
- Cuvettes
- Pipettes and tips
- Gloves, lab coat, protective eyewear

5.2 Safety Notes

Always wear gloves and handle blood as a biohazard.

Drabkin's reagent contains potassium cyanide, a highly potent neurotoxin. Prevent inhalation or other absorption of the substance. Be mindful of proper waste disposal.

5.3 Steps

1. Prepare sample and reference cuvettes.
 1. Fill both cuvettes with 5mL of Drabkin's reagent.
 2. Add 20 μ L of blood to the sample cuvette.
2. Gently shake the sample cuvette to mix the contents.
3. Incubate for ≥ 5 minutes in the dark for full conversion to cyan-methemoglobin HiCN.
4. Set the spectrophotometer to 546 nm.
5. Zero the device using the reference cuvette.
6. Insert the sample cuvette and record the absorbance ($E_{546} = A$).
7. Calculate the hemoglobin concentration using Equation 13.

5.4 Calculation of Hemoglobin Concentration

5.4.1 Seemingly wrong formula

Using the Beer-Lambert law from Section 1.2.1 we formulate:

$$(10) \quad c [\text{mol/L}] = \frac{A}{\varepsilon} \times \frac{1}{\ell}$$

We can calculate our absorptivity ε of HiCN for the total sample:

$$\begin{aligned}
 (11) \quad \epsilon_{\text{total}} &= \frac{\epsilon_{\text{HiCN}}}{\text{MG}_{\text{Hb}}} \times \frac{V_{\text{probe}}}{V_{\text{total}}} \\
 &= \frac{44\,000 \frac{\text{L}}{\text{mol} \cdot \text{cm}}}{64\,458 \frac{\text{g}}{\text{mol}}} \times \frac{20\,\mu\text{L}}{5.02\,\text{mL}} \\
 &\approx 0.272 \frac{\text{m}^2}{\text{kg}} = 2.72 \frac{\text{L}}{\text{kg} \cdot \text{cm}}
 \end{aligned}$$

This was calculated using the following values:

- molar absorptivity of HiCN: $\epsilon_{\text{HiCN}} = 44\,000 \frac{\text{L}}{\text{mol} \cdot \text{cm}}$
- molar mass of Hb: $\text{MG}_{\text{Hb}} = 64\,458 \frac{\text{g}}{\text{mol}}$
- volume of the blood probe: $V_{\text{probe}} = 20\,\mu\text{L}$
- volume of the total sample: $V_{\text{total}} = 5.02\,\text{mL}$

Since we know that our cuvette has a thickness of $\ell = 1\,\text{cm}$ we can calculate the concentration of Hb in the sample:

$$\begin{aligned}
 (12) \quad c \text{ [kg/L]} &= \frac{A}{\epsilon_{\text{total}}} \times \frac{1}{\ell} \\
 &= \frac{A}{2.72 \frac{\text{L}}{\text{kg} \cdot \text{cm}}} \times \frac{1}{0.01\,\text{m}} \\
 &= A \times \frac{1}{2.72 \frac{\text{L}}{\text{kg} \cdot \text{cm}} \times 1\,\text{cm}} \\
 &= A \times \frac{1}{2.72} \frac{\text{kg}}{\text{L}} \\
 &\approx A \times 0.368 \frac{\text{kg}}{\text{L}} \\
 \Rightarrow c \text{ [g/dL]} &= A \times 36.77 \frac{\text{g}}{\text{dL}}
 \end{aligned}$$

5.4.2 Correct formula from instructor

The instructor provided us with a formula containing the precalculated factor to calculate the hemoglobin concentration:

$$(13) \quad c \text{ [g/dL]} = A \times 14.746 \frac{\text{g}}{\text{dL}}$$

We will use this factor in all further applications, even though there seems to be no explanation for the significant difference between the two formulas. Plugging in the values in their formula $A \times \frac{\text{MG}_{\text{Hb}}}{\epsilon_{\text{HiCN}}} \times \frac{V_{\text{total}}}{V_{\text{probe}}} \times \frac{1}{d \text{ [L]}}$ does not yield the same result for me.

Example calculation using placeholder values:

$$A = 1.21$$

$$c_{\text{Hb}} = 1.21 \times 14.746 \frac{\text{g}}{\text{dL}} = 17.843 \frac{\text{g}}{\text{dL}}$$

6 Results

We analyzed the data from

1. our class
2. the data of all years, ranging from 2015 to 2024

6.1 Our class: MBI24 (current year)

Table 3 shows the summary of our class. In total 22 students participated in the experiment, split between 12 females and 10 males. One can observe that the mean c_{Hb} of the females is lower than the mean of the males, and the range also suggests that the males hemoglobin concentration tends to be higher.

Group	Amount	Range ($\frac{g}{dL}$)	Mean $\pm$ SD ($\frac{g}{dL}$)	Variance
Male	10	10.9 – 23.0	16.7 $\pm$ 3.5	13.5
Female	12	10.1 – 17.3	12.1 $\pm$ 4.3	19.9

Table 3: MBI24 results summary male vs. female

We expect the difference in c_{Hb} between males and females to be statistically significant. Therefore we performed a two-sample t-test with different variances using an α of 5% to test the null hypothesis that the difference in c_{Hb} between males and females is zero.

The results of the test are shown in Table 4. The t-statistic is 2.63, the critical t-value is 1.72, the p-value is 0.81% and the Cohen's d is 0.32.

Since the p-value is less than the chosen $\alpha = 5\%$, we reject the null hypothesis and conclude that the difference in c_{Hb} between males and females is statistically significant. This is also supported by the t-statistic of 2.63 exceeding the critical t-value of 1.72.

Key Values	
chosen α	5%
t-statistic	2.63
critical t-value	1.72
p-value	0.81%
Cohen's d	0.32

Table 4: MBI24 t-test values

The Cohen's d of 0.32 indicates a small effect size, which means that the difference in c_{Hb} between males and females is not very large.

6.2 All years: 2015 - 2024

The analysis is repeated for the years 2015 to 2024 (including MBI24). The results are shown in Table 5. We can observe the same tendency as in Section 6.1 of a higher c_{Hb} in males than in females.

Group	Amount	Range ($\frac{g}{dL}$)	Mean $\pm$ SD ($\frac{g}{dL}$)	Variance
Male	117	2.32 – 53.2	16.72 $\pm$ 7.83	61.36
Female	126	0.37 – 59.8	14.87 $\pm$ 7.24	52.35

Table 5: All years results summary male vs. female

For this dataset we also performed a two-sample t-test with different variances using an α of 5%.

The results are shown in Table 6. The t-statistic is 1.66, the critical t-value is 1.65, the p-value is 4.98% and the Cohen's d is 0.026.

Since the p-value is less than the chosen $\alpha = 5\%$, we reject the null hypothesis and conclude that the difference in c_{Hb} between males and females is statistically significant. This is also supported by the t-statistic of 1.66 being slightly greater than the critical t-value of 1.65.

The Cohen's d of 0.026 indicates a very small effect size, which means that the difference in c_{Hb} between males and females is tiny.

Key Values	
chosen α	5%
t-statistic	1.66
critical t-value	1.65
p-value	4.98%
Cohen's d	0.026

Table 6: All years t-test values

This matches the intuitive interpretation of the data based on the boxplots seen in, where the observed difference is rather marginal.

6.3 Conclusion

The results of both t-tests show that the difference in c_{Hb} between males and females is statistically significant. This is also supported by the boxplots and the key values. We can observe less pronounced differences in the all years dataset, which can probably be attributed to the higher sample size and therefore more noise accumulation.

In Section 6.1 covering the class of the current year (MBI24), we can observe a significant difference in c_{Hb} between males and females, despite the comparatively small sample size. This indicates high accuracy of the hemiglobincyanide method.

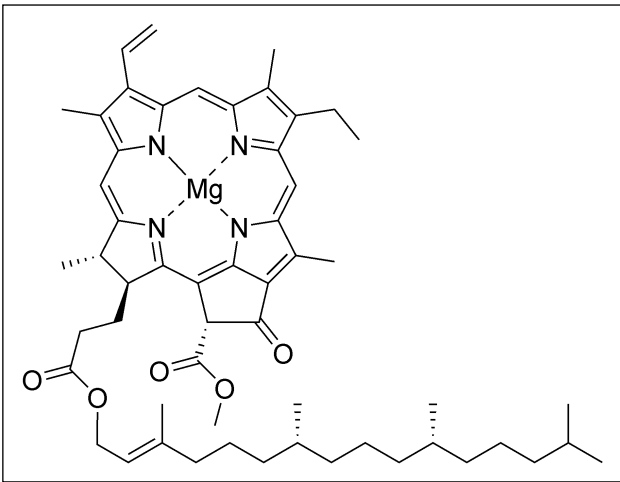


Figure 5: Chlorophyll A structure ^[1]

("800", "-0.054")

Appendix

Sources

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